MATH 110 — Week 7

Notes Chapters 12–13; Homework 7

§1 Upper-triangular matrices

The question. How does a matrix record a nested family of invariant subspaces? For $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$, the first coordinate line is invariant because $Ae_1 = e_1$.

Tools from earlier weeks.

- Week 2: extend an independent list to a basis by adding vectors outside its span; basis length defines dimension and an independent list of full length is a basis.
- Weeks 3–4: rank-nullity follows by extending a kernel basis and using the remaining images as a range basis; thus an operator on finite dimension is injective, surjective and invertible under the same conditions.
- Week 6: for $p(x) = \sum_j a_j x^j$, define $p(T) = \sum_j a_j T^j$ with $T^0 = I$. Eigenvectors for distinct eigenvalues are independent by applying a shifted operator to a shortest dependence; polynomial evaluation preserves products by distributing finite sums of powers, so polynomials in T commute.
- Weeks 4–6: the root bound follows by successively removing root factors; $\dim(V/U) = \dim V - \dim U$ follows by extending a basis of U: the added vectors give a quotient basis; evaluating a polynomial on T/U sends $v + U$ to $p(T)v + U$, by induction on powers.
- Week 6: a probability column has nonnegative real entries summing to 1; a column-stochastic matrix has nonnegative entries with each column summing to 1. With $\|x\|_1 = \sum_i |x_i|$, such a P satisfy $\|Px\|_1 \leq \sum_{i,j} P_{ij}|x_j| = \|x\|_1$, by the triangle inequality and column sums 1.
- Week 6: every operator on a nonzero finite-dimensional complex space has an eigenvalue by factoring a relation among vector iterates and using the fundamental theorem of algebra. That theorem was proved there using the explicitly cited extreme-value theorem; its repeated root extraction gives complex factorization.

Definition (Notes Definition 33; LADR 5.37–5.38, 5.48)**.** Work with linear $T : V \to V$ on finite-dimensional V over F, unless otherwise specified. A square matrix $A = (a_{ij})$ is upper triangular when $a_{ij} = 0$ for $i > j$, and diagonal when $a_{ij} = 0$ for $i \neq j$. The main diagonal consists of the entries a_{jj}.

Example (Notes Example 53)

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}, \qquad Tv_3 = 3v_1 + 6v_2 + 8v_3.$$

Column 3 uses only the first three basis vectors; its fourth entry is zero. All entries below the diagonal are zero, so A is upper triangular with diagonal $1, 5, 8, 10$. It is not diagonal, since $a_{12} = 2 \neq 0$.

Proposition (Notes Proposition 59; LADR 5.39)

For a basis $v_1, \dots, v_n$, the following are equivalent: the matrix of T is upper triangular; $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j; every initial span $U_j = \operatorname{span}(v_1, \dots, v_j)$ is invariant.

Proof. 1. Column j means $Tv_j = \sum_i a_{ij} v_i$. Thus $a_{ij} = 0$ for $i > j$ iff $Tv_j \in U_j$, proving the first two conditions equivalent.
2. If $Tv_j \in U_j$, then for $k \le j$, $Tv_k \in U_k \subseteq U_j$; linearity proves $T(U_j) \subseteq U_j$.
3. If U_j is invariant, $v_j \in U_j$ gives $Tv_j \in U_j$.

□

Proposition (Notes Proposition 60; LADR 5.41)

The eigenvalues of an upper-triangular matrix are exactly its diagonal entries, viewed as elements of the underlying field.

Proof. 1. For an upper-triangular B with every diagonal entry nonzero, solve $Bx = y$ from the last row upward: each row determines its new coordinate uniquely by division by b_{jj}. Thus B is invertible.
2. If some diagonal entry is zero, let j be the first such index. Put $x_j = 1$ and $x_k = 0$ for $k > j$; solve the first $j - 1$ equations upward in reverse order using their nonzero diagonal entries. Then $Bx = 0$ with $x \neq 0$.
3. Apply these facts to $B = A - \lambda I$, whose diagonal entries are $a_{jj} - \lambda$.

□

Example (Notes Example 54) 1. On $\mathcal{P}_3(F)$ for $F = \mathbb{R}$ or $\mathbb{C}$, the power rule gives differentiation matrix $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ in $1, x, x^2, x^3$.
2. Its only eigenvalue is 0 by Proposition 60. Its eigenspace is the constants because $Dp = 0$ exactly when every positive-degree coefficient is zero.

$$x^3 \xrightarrow{D} 3x^2 \xrightarrow{D} 6x \xrightarrow{D} 6 \xrightarrow{D} 0$$

Differentiation sends each polynomial to a lower-degree initial span.

Proposition (Notes Proposition 61; LADR 5.47)

Every operator on a finite-dimensional complex vector space has an upper-triangular matrix in some basis.

Where the proof comes from

Want. Construct a basis with invariant initial spans.

Know. Every nonzero finite-dimensional complex space has an eigenvector (Week 6).

Key step. We need a smaller invariant space whose triangular basis will also control the added vectors. Choose $U = \operatorname{range}(T - \lambda I)$: the eigenvalue makes it proper, and $Tv - \lambda v \in U$ for every v. Thus each added basis vector maps into U plus its own line, exactly the required column condition.

Proof. 1. Induct on $n = \dim V$; the empty basis handles $n = 0$. For $n > 0$, choose an eigenvalue λ by Week 6.
2. Put $U = \operatorname{range}(T - \lambda I)$. It is proper because $T - \lambda I$ is not surjective; it is invariant because it commutes with T.
3. By induction, take a triangular basis $v_1, \dots, v_m$ for $T|_U$, then extend it to a basis of V (Week 2).
4. For $j > m$, $Tv_j = (T - \lambda I)v_j + \lambda v_j \in U + Fv_j \subseteq U_j$. For $j \le m$ the induction hypothesis gives the same containment. Apply Proposition 59.

□

Shortcuts

- Triangular eigenvalues are the diagonal entries, because subtracting λI changes only that diagonal.
- To prove triangular form, check initial-span invariance on basis vectors.

Traps

- A triangular matrix need not have a basis of eigenvectors: $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has only the eigenvector line Fe_1.
- Real triangularization can fail: the real quarter-turn has no real eigenvalue, whereas every nonempty real triangular matrix has a diagonal eigenvalue.

§2 Diagonalization and repeated application

The question. When can repeated application be reduced to scalar powers? For $\operatorname{diag}(2,3)$, its kth power sends (a,b) to $(2^k a, 3^k b)$.

Definition (Notes Definition 34; LADR 5.50, 5.52)**.** An operator is diagonalizable over F if its matrix in some basis is diagonal. An eigenbasis is a basis of eigenvectors. Write $E(\lambda, T) = \ker(T - \lambda I) = \{v : Tv = \lambda v\}$ (Week 6); this is $\{0\}$ when λ is not an eigenvalue. Its dimension is the geometric multiplicity of λ.

Proposition (Notes Proposition 62; LADR 5.54–5.55)

For any distinct eigenvalues, their eigenspace sum is direct and $\sum_j \dim E(\lambda_j, T) \le \dim V$. For the complete list of distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an operator on finite-dimensional V, the following are equivalent: T is diagonalizable; V has an eigenbasis; $V = \bigoplus_{j=1}^{r} E(\lambda_j, T)$; $\dim V = \sum_j \dim E(\lambda_j, T)$.

Where the proof comes from

Want. Decide whether the available eigenspaces provide a full basis.

Know. Eigenvectors with distinct eigenvalues are independent (Week 6).

Key step. Group a relation by eigenvalue. Each nonzero grouped sum is itself an eigenvector; distinct-eigenvalue independence makes each group zero. A basis in each eigenspace then gives one independent list, and counting its length decides whether it fills V.

Proof. 1. If $w_j \in E(\lambda_j, T)$ and $\sum_j w_j = 0$, the nonzero w_j are eigenvectors with distinct eigenvalues; Week 6 forces every $w_j = 0$. Thus the sum is direct. Its dimension is the sum of the eigenspace dimensions and is at most $\dim V$, by Week 4.

2. A diagonal matrix has $Tv_j = \lambda_j v_j$ for each basis column, and an eigenbasis has a diagonal matrix. This proves the first two conditions equivalent.

3. Concatenating bases of the distinct eigenspaces gives an independent list by step 1. It is a basis precisely when the eigenspaces span V, equivalently when its length equals $\dim V$ (Week 2).

□

Example (LADR 5.49, 5.51, 5.53) 1. A diagonal matrix $\operatorname{diag}(8,5,5)$ in basis v_1, v_2, v_3 has eigenspaces $E(8,T) = Fv_1$ and $E(5,T) = \operatorname{span}(v_2, v_3)$ over $\mathbb{R}$ or $\mathbb{C}$, by solving the coordinate equations.

2. For $A = \left(\begin{smallmatrix} 41 & 7 \\ -20 & 74 \end{smallmatrix}\right)$, direct multiplication gives $A(1,4) = 69(1,4)$ and $A(7,5) = 46(7,5)$.

3. The vectors are independent since $a(1,4) + b(7,5) = 0$ gives $a = -7b$ and $-23b = 0$, hence $a = b = 0$ over $\mathbb{R}$. Thus its matrix in that basis is $\operatorname{diag}(69, 46)$.

Proposition (Notes Proposition 63; LADR 5.58)

An operator on an n-dimensional space with n distinct eigenvalues is diagonalizable.

Proof. 1. Choose $v_j \neq 0$ with $Tv_j = \lambda_j v_j$. Week 6 gives independence.
2. An independent list of length n is a basis (Week 2), hence an eigenbasis. The empty list handles $n = 0$.

□

Example (Notes Example 55; LADR 5.57)

For the textbook case $T(a, b, c) = (b, c, 0)$, the standard matrix is $\left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. More generally, for $n > 1$, let $Tv_1 = 0$ and $Tv_j = v_{j-1}$ for $j > 1$.

1. The matrix is upper triangular with zero diagonal, so its only eigenvalue is 0 by Proposition 60.
2. Solving $T\sum_j a_j v_j = \sum_{j>1} a_j v_{j-1} = 0$ gives $E(0, T) = Fv_1$, of dimension 1.
3. Since $1 < n$, Proposition 62 rules out diagonalizability.

Example (Notes Example 56) 1. For $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, the Week 6 two-by-two test gives $(t-1)(t-3)-4 = t^2-4t-1$, with roots $2 \pm \sqrt{5}$. Each has eigenvector $(2, 1 \pm \sqrt{5})$, verified by multiplication.
2. Thus A is diagonalizable over $\mathbb{R}$ by Proposition 63. Over $\mathbb{Q}$ it has no eigenvalue, since the roots are irrational, so it has no eigenbasis.
3. For $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the eigenvalues over $\mathbb{C}$ are $i, -i$ with vectors $(1, -i), (1, i)$; over $\mathbb{R}$ there is no eigenvalue.

Proposition (Notes Proposition 64)

If $Tv_j = \lambda_j v_j$ is an eigenbasis and $v = \sum_j a_j v_j$, then $T^k v = \sum_j a_j \lambda_j^k v_j$ for every integer $k \geq 0$. Here $\lambda_j^0 = 1$, including when $\lambda_j = 0$.

Proof. 1. Induction on k gives $T^k v_j = \lambda_j^k v_j$, with $k = 0$ the identity equation.
2. Apply T^k to the finite linear combination for v and distribute by linearity.

□

Compute a large power (LADR 5.59) Over $\mathbb{R}$ or $\mathbb{C}$, let $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{smallmatrix}\right)$.

1. In $(A - \lambda I)(x, y, z)^{\mathsf{T}} = 0$, the rows give $(2-\lambda)x + y = 0$, $(5-\lambda)y + 3z = 0$, $(8-\lambda)z = 0$. For $\lambda = 2$ these force $z = y = 0$; for 5, $z = 0, y = 3x$; for 8, $y = z = 6x$. Choose $x = 1$ to obtain $v_1 = (1, 0, 0)$, $v_2 = (1, 3, 0)$, $v_3 = (1, 6, 6)$.
2. For $e_3 = av_1 + bv_2 + cv_3$, the equations are $a + b + c = 0$, $3b + 6c = 0$, $6c = 1$. Hence $(a, b, c) = (1/6, -1/3, 1/6)$.
3. Proposition 64 gives $A^{100}e_3 = \frac{1}{6}(2^{100} - 2 \cdot 5^{100} + 8^{100},\ 6 \cdot 8^{100} - 6 \cdot 5^{100},\ 6 \cdot 8^{100})$.

Recover the Fibonacci formula (Notes Chapter 12; LADR 5D Exercise 21)

1. Let V be the real sequences satisfying $a_{n+2} = a_{n+1} + a_n$ for $n \geq 0$. Sending a sequence to (a_0, a_1) is an isomorphism: the recurrence determines every later term uniquely.
2. The left shift corresponds to $T(x, y) = (y, x + y)$. Its eigenvalues are $\lambda_\pm = (1 \pm \sqrt{5})/2$, with eigenvectors $(1, \lambda_\pm)$ because $\lambda_\pm^2 = \lambda_\pm + 1$.
3. For the Fibonacci start $(F_0, F_1) = (0, 1)$, $(0, 1) = ((1, \lambda_+) - (1, \lambda_-))/\sqrt{5}$. The first coordinate after n shifts is therefore $F_n = (\lambda_+^n - \lambda_-^n)/\sqrt{5}$.

Shortcuts

- Solve $Sc = v$ once, where the columns of S are eigenvectors; then replace each c_j by $\lambda_j^k c_j$.
- Distinct eigenvalues certify diagonalizability before the eigenvectors are computed.

Traps

- A repeated eigenvalue may still allow diagonalization: I_2 has only eigenvalue 1 and every basis is an eigenbasis.
- The field matters: the quarter-turn is diagonalizable over $\mathbb{C}$, but has no real eigenbasis.

§3 Stochastic powers and convergence rates

The question. What determines the rate of approach to a stationary distribution? For $P_\epsilon = \left(\begin{smallmatrix} 1-\epsilon & \epsilon \\ \epsilon & 1-\epsilon \end{smallmatrix}\right)$, the difference between the two coordinates is multiplied by $1-2\epsilon$ at each step.

Definition (Notes Chapter 12: eigenmodes and spectral gap)**.** In an eigenbasis expansion, an eigenmode means one component $a_j v_j$. Over $\mathbb{R}$ or $\mathbb{C}$, its coefficient after k iterations has modulus $|a_j|\,|\lambda_j|^k$ by Proposition 64 and multiplicativity of modulus. For nonzero a_j, this tends to zero if $|\lambda_j| < 1$, is constant if $|\lambda_j| = 1$, and tends to infinity if $|\lambda_j| > 1$, by the geometric limits from Week 6 (use reciprocal for growth). For a stochastic operator with a single stationary direction and other eigenvalues of modulus less than 1, its spectral gap is $1 - \max_{\lambda \neq 1} |\lambda|$. If there are no other eigenvalues, take this maximum to be 0.

Example (Notes Chapter 12: the three-state chain) 1. Let $P = \left(\begin{smallmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{smallmatrix}\right)$, $\pi = (1/4, 1/2, 1/4)$, $u = (-1, 0, 1)$ and $w = (1, -2, 1)$. Multiplication gives $P\pi = \pi$, $Pu = u/2$ and $Pw = 0$.

2. These vectors form an eigenbasis because their eigenvalues $1, 1/2, 0$ are distinct. Solving for the initial distribution gives $e_1 = \pi - u/2 + w/4$.
3. For $k \geq 1$, Proposition 64 gives $P^k e_1 = \pi - 2^{-(k+1)} u = (1/4 + 2^{-(k+1)}, 1/2, 1/4 - 2^{-(k+1)})$.
4. The stationary component stays fixed, the 0-component becomes zero after one step, and the remaining error is halved each step. In particular $\|P^k e_1 - \pi\|_1 = 2^{-k}$ for $k \geq 1$.

Example (Notes Chapter 12: a bottleneck) 1. For $0 < \epsilon < 1/2$, $P_\epsilon(1,1) = (1,1)$ and $P_\epsilon(1,-1) = (1-2\epsilon)(1,-1)$.

2. Since $e_1 = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$, its kth distribution is $\frac{1}{2}(1,1) + \frac{1}{2}(1-2\epsilon)^k(1,-1)$.
3. Its gap is 2ϵ. Small ϵ makes transitions between the two states rare and the error factor $1-2\epsilon$ close to 1, hence slow convergence.

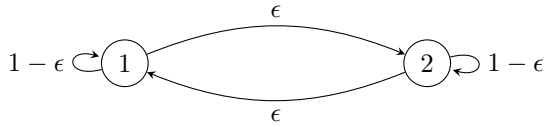
Proposition (Notes Chapter 12: PageRank eigenvalues)

Let P be column-stochastic, v a probability column and $0 < \alpha < 1$. Put $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$ and $G = \alpha P + (1-\alpha) v \mathbf{1}^\mathsf{T}$. For the unique PageRank probability vector r from Week 6, $\mathbb{R}^n = \mathbb{R}r \oplus H$, where $H = \{x : \mathbf{1}^\mathsf{T} x = 0\}$, $Gr = r$, and $G|_H = \alpha P|_H$. Every complex eigenvalue of the second restriction has modulus at most α.

Proof. 1. Column sums give $\mathbf{1}^\mathsf{T} P = \mathbf{1}^\mathsf{T}$. Thus H is invariant and $Gx = \alpha P x$ on H, while $Gr = r$. Every $x = (\mathbf{1}^\mathsf{T} x) r + [x - (\mathbf{1}^\mathsf{T} x) r]$ has its bracket in H; $cr \in H$ forces $c = 0$, proving directness. These equations also hold over $\mathbb{C}$ with the complex coordinate-sum-zero subspace.

2. If $Gx = \mu x$ for nonzero $x \in H$, then $|\mu| \|x\|_1 = \|Gx\|_1 = \alpha \|Px\|_1 \leq \alpha \|x\|_1$. Divide by $\|x\|_1 > 0$ to obtain $|\mu| \leq \alpha$.
3. For probability starts, Week 6 already gives $\|r_k - r\|_1 \leq \alpha^k \|r_0 - r\|_1$ directly; no eigenbasis assumption is required.

□



The two-state matrix has probability $1-\epsilon$ of remaining in its current state when ϵ is small. Its error factor is $1-2\epsilon$.

Shortcuts

- Subtract the stationary vector first: only the other eigenmodes remain, so their moduli determine the error rates when an eigenbasis exists.
- The PageRank 1-norm contraction bound remains valid without diagonalization, by Week 6.

Traps

- Stochasticity alone does not imply convergence: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ alternates e_1 and e_2.
- A small gap means slow convergence: for $\epsilon = 1/100$, the nonstationary factor is $49/50$, not $1/50$.

§4 The minimal polynomial

The question. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $A^2 = 4A - 3I$. How can we find the first such relation, determine the eigenvalues, and reduce every higher power to a combination of I, A?

Example (LADR 5.20: finite dimension is necessary) 1. On $\mathcal{P}(\mathbb{C})$, define $(Tp)(x) = xp(x)$. This is linear by distributivity of polynomial multiplication.

2. For $p \neq 0$, $\deg(Tp) = \deg p + 1$, whereas a nonzero scalar multiple of p has degree $\deg p$ and $0p = 0 \neq Tp$. Thus T has no eigenvalue. The space is infinite-dimensional because $1, x, x^2, \ldots$ is independent by coefficient comparison.

Definition (Notes Definition 35; LADR 5.14)**.** For $p(x) = \sum_j a_j x^j$, put $p(T) = \sum_j a_j T^j$, with $T^0 = I$. Week 6 proved $(pq)(T) = p(T)q(T)$, linearity of evaluation, and $p(T)v = p(\lambda)v$ when $Tv = \lambda v$.

Definition (Notes Definition 36; LADR 5.21, 5.24)**.** A polynomial is monic if its leading coefficient is 1. For a finite-dimensional operator, its minimal polynomial m_T is the monic polynomial of smallest degree satisfying $m_T(T) = 0$. A nonzero polynomial p satisfying $p(T) = 0$ is called an annihilating polynomial. A polynomial q is divisible by p if $q = ph$ for a polynomial h. A monic polynomial splits over F if it is a product of linear factors $x - \lambda_j$ with $\lambda_j \in F$. On the zero space, $I = 0$ and $m_T = 1$.

Proposition (Existence and uniqueness in Notes Definition 36; LADR 5.22)
The minimal polynomial exists and is unique. If $I, T, \ldots, T^{d-1}$ are independent and $T^d = \sum_{j<d} a_j T^j$, then $m_T(x) = x^d - \sum_{j<d} a_j x^j$.

Where the proof comes from

Want. Obtain a first polynomial relation and make it unique.

Know. Week 3 identifies $L(V)$ with $n \times n$ matrices, $n = \dim V$; the n^2 single-entry matrices form a basis, so $\dim L(V) = n^2$.

Key step. The first dependence has a nonzero coefficient on its last power; divide by it to solve that power in terms of the independent earlier powers. This produces a monic relation, and subtracting two monic relations cancels the last power, which explains uniqueness.

Proof. 1. The list $I, T, T^2, \ldots$ is dependent in $L(V)$. A nonzero relation gives an annihilating polynomial; choose its smallest possible degree and divide by its leading coefficient.

2. Two distinct monic polynomials of that degree would have a nonzero difference of smaller degree, also annihilating T, contradicting minimality.

3. Independence of the earlier powers rules out any smaller-degree relation, proving the stated formula. The case $V = \{0\}$ gives the constant polynomial 1.

□

Example (Notes Example 57) 1. On nonzero F^2, $m_I = x - 1$, since $I - I = 0$ and no nonzero constant annihilates I.

2. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $A^2 = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right) = 4A - 3I$. Thus $x^2 - 4x + 3$ annihilates A.

3. The matrices I, A are independent because A is not scalar; hence $m_A = x^2 - 4x + 3 = (x - 1)(x - 3)$.

Proposition (Polynomial division; LADR 4.9)

If $p \neq 0$, every polynomial q has a unique expression $q = ph + r$, where $r = 0$ or $\deg r < \deg p$.

Where the proof comes from

Want. Reduce a polynomial relation to a smaller degree.
Know. Leading coefficients can be divided in a field.
Key step. Repeatedly cancel the leading term; the degree drops each time.

Proof. 1. If $\deg q \geq \deg p$, subtract the unique scalar multiple of $x^{\deg q - \deg p} p$ that cancels its leading term. Record that monomial in h.
2. Repeat until the remainder is zero or has smaller degree than p. The decreasing nonnegative degrees ensure termination.
3. If $ph + r = ph' + r'$, then $p(h - h') = r' - r$. A nonzero left side has degree at least $\deg p$, while the right side has smaller degree; both sides must be zero.

□

Proposition (Notes Proposition 65; LADR 5.29)

For every polynomial q, $q(T) = 0$ if and only if m_T divides q.

Proof. 1. If $q = m_T h$, Week 6 polynomial multiplication gives $q(T) = m_T(T)h(T) = 0$.
2. Conversely, write $q = m_T h + r$. If $q(T) = 0$, then $r(T) = 0$; a nonzero r normalized to be monic would contradict minimality. Thus $r = 0$.

□

Proposition (Relations on one vector; Midterm 2 Review, Problem 6)

For $v \in V$ there is a unique monic polynomial p_v of least degree with $p_v(T)v = 0$. For every polynomial q,

$$q(T)v = 0 \iff p_v \mid q; \qquad p_v \mid m_T.$$

For $v = 0$, $p_v = 1$.

Proof. 1. For $v \neq 0$, the first dependence among $v, Tv, \ldots$ gives a relation; divide by its last nonzero coefficient to make it monic. Two least-degree monic choices have a smaller-degree difference with zero value on v, so they agree.
2. Divide $q = p_v h + r$, with $r = 0$ or $\deg r < \deg p_v$. If $q(T)v = 0$, commutation gives $r(T)v = q(T)v - h(T)p_v(T)v = 0$. Minimality forces $r = 0$. Conversely, a multiple of p_v has zero value on v by the same product equation.
3. Apply this to $q = m_T$. For $v = 0$, every polynomial gives zero, and the unique monic degree-zero choice is 1.

□

Definition (Least common multiple)**.** The monic least common multiple $\operatorname{lcm}(p_1, \ldots, p_r)$ of nonzero polynomials is a monic common multiple that divides every common multiple.

1. The product is a common multiple, so choose a monic common multiple L of least degree.
2. Divide any common multiple $q = Lh + r$. Every p_j divides q and Lh, hence divides r. A nonzero r would be a common multiple of smaller degree; thus $r = 0$ and $L \mid q$.
3. Two such monic choices divide each other and therefore agree. This proves existence and uniqueness.

Proposition (Combine relations on a basis)

For any basis $v_1, \dots, v_n$,

$$m_T = \operatorname{lcm}(p_{v_1}, \dots, p_{v_n}).$$

The empty least common multiple is 1 on $V = \{0\}$.

Proof. 1. A linear map $q(T)$ is zero iff it is zero on every basis vector: expand $q(T)(\sum_j a_j v_j) = \sum_j a_j q(T) v_j$.

2. By the vector relation criterion, this holds iff every p_{v_j} divides q. Their least common multiple and m_T therefore have exactly the same polynomial multiples; each divides the other and both are monic, so they agree.

□

Proposition (Notes Proposition 66; LADR 5.27(a)–(b))

The eigenvalues of T are exactly the roots of m_T in the underlying field. Over $\mathbb{C}$, $m_T = \prod_j (x - \lambda_j)^{k_j}$ for the distinct eigenvalues λ_j and positive integers k_j (the zero-space product is empty).

Where the proof comes from

Want. Recover eigenvalues from a polynomial relation of minimal degree.

Know. Polynomials act on eigenvectors by scalar evaluation; roots give linear factors (Week 4).

Key step. Factor $m_T = (x-\lambda)q$. Minimality forces $q(T) \neq 0$; choose w with $q(T)w \neq 0$. Then $(T-\lambda I)q(T)w = 0$, explicitly producing an eigenvector.

Proof. 1. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$.

2. If $m_T(\lambda) = 0$, factor $m_T = (x - \lambda)q$. The smaller degree of q forces $q(T) \neq 0$ by minimality. Hence some w has $u = q(T)w \neq 0$, and $(T - \lambda I)u = m_T(T)w = 0$ makes u an eigenvector.

3. For a matrix, any nonzero column of $q(A)$ gives such a vector: it is $q(A)e_j$. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $\lambda = 3$, $q(A) = A - I = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ gives eigenvector $(1, 1)$.

4. Over $\mathbb{C}$, factor m_T by the fundamental theorem of algebra from Week 6, whose analytic input was the cited extreme-value theorem. The preceding equivalence identifies every root as an eigenvalue, giving the product.

□

Proposition (Relations on invariant parts; LADR 5.31 and quotient analogue)

For invariant $U \subseteq V$, put $r = m_{T|_U}$ and $s = m_{T/U}$. Then

$$r \mid m_T, \qquad s \mid m_T, \qquad m_T \mid rs.$$

Where the proof comes from

Want. Combine a relation valid only on U with one valid on the quotient.

Key step. $s(T/U) = 0$ says $s(T)v + U = U$, so $s(T)$ sends every vector into U. On U, $r(T)$ sends every vector to zero. Thus $r(T)s(T) = 0$ on all of V. Restricting a global zero relation, or applying it to cosets, gives the opposite divisibility statements.

Proof. 1. For $u \in U$, $m_T(T|_U)u = m_T(T)u = 0$. On a coset, $m_T(T/U)(v + U) = m_T(T)v + U = U$. Proposition 65 on the two spaces gives $r, s \mid m_T$.

2. For each v, $s(T)v \in U$ and then $r(T)s(T)v = 0$. Thus $(rs)(T) = 0$; Proposition 65 gives $m_T \mid rs$.

□

Proposition (Notes Proposition 67; LADR 5.22)

For finite-dimensional V, $\deg m_T \leq \dim V$.

Where the proof comes from

Want. Improve the operator-space bound $(\dim V)^2$ to $\dim V$.

Know. Vector iterates become dependent within $\dim V + 1$ terms, and invariant subspaces have quotient operators (Week 5).

First try. A relation on one vector need not hold on all vectors: $(T - I)e_1 = 0$ for $T = \operatorname{diag}(1, 2)$, but $T - I \neq 0$.

Key step. Put every vector satisfying the chosen relation into $U = \ker q(T)$, which contains v and is invariant. If $U \neq V$, both U and V/U are smaller. Their relations combine by the preceding result: the quotient relation sends vectors into U, then the restriction relation sends those images to zero. Their degree bounds add to $\dim U + \dim(V/U) = \dim V$.

Proof. 1. Induct on $n = \dim V$; for $n = 0$, $m_T = 1$ has degree 0. For $n > 0$, choose $v \neq 0$. A dependence of $v, Tv, \dots, T^n v$ gives nonzero q of degree at most n with $q(T)v = 0$.

2. Set $U = \ker q(T)$, a nonzero invariant subspace by Week 6. If $U = V$, Proposition 65 gives $m_T \mid q$, proving the bound.
3. Otherwise, $0 < \dim U < n$. Induction gives minimal polynomials r for $T|_U$ and s for T/U with $\deg r \leq \dim U$, $\deg s \leq \dim(V/U)$.
4. The relations-on-parts result gives $m_T \mid rs$: the quotient relation sends vectors into U, then the restriction relation sends those images to zero.
5. Consequently $\deg m_T \leq \deg r + \deg s \leq \dim U + \dim(V/U) = n$, by Week 5.

□

Example (Notes Example 58) 1. On any nonzero space the identity has minimal polynomial $x - 1$.

2. For the length-n shift of Example 55, $T^n = 0$ but $T^{n-1}v_n = v_1 \neq 0$. In fact $v_n, Tv_n, \dots, T^{n-1}v_n$ is a basis, so no polynomial of degree below n annihilates T; therefore $m_T = x^n$.
3. For each $1 \leq d \leq n$, let T be a length-d shift on $v_1, \dots, v_d$ and zero on the remaining basis vectors. Then $T^d = 0$, while $v_d, Tv_d, \dots, T^{d-1}v_d$ are independent. Thus $m_T = x^d$, realizing every degree. For $n = 0$, $m_T = 1$ has degree 0.

Find the first relation among matrices

1. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, try $A = aI$: the off-diagonal entry 1 makes this impossible.
2. Next solve $A^2 = aI + bA$ entrywise. Off-diagonal entries give $b = 4$, then diagonal entries give $a = -3$. The first relation yields $m_A = x^2 - 4x + 3$.

Definition (LADR 5B: cyclic vector)**.** A vector v is cyclic for T on an n-dimensional space if $v, Tv, \dots, T^{n-1}v$ is a basis.

Proposition (LADR 5B: cyclic-vector computation)

If $v, Tv, \dots, T^{n-1}v$ is a basis and $T^n v = -\sum_{j<n} a_j T^j v$, then $m_T = x^n + \sum_{j<n} a_j x^j$.

Proof. 1. For the displayed polynomial p, $p(T)v = 0$, hence $p(T)T^j v = T^j p(T)v = 0$ for every basis vector. Thus $p(T) = 0$.

2. No lower-degree nonzero polynomial can annihilate v, by independence of the basis iterates. Therefore no such polynomial annihilates T and $p = m_T$.

□

Example (LADR 5.26, 5.28)

Over $\mathbb{C}$, let $C = \begin{pmatrix} 0&0&0&0&-3 \\ 1&0&0&0&6 \\ 0&1&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&1&0 \end{pmatrix}$, $e_1, Ce_1, \ldots, C^4e_1 = e_1, e_2, \ldots, e_5$ and $C^5e_1 = -3e_1 + 6Ce_1$. The cyclic computation gives $m_C = x^5 - 6x + 3$. Its eigenvalues are its roots by Proposition 66, approximately -1.67, 0.51, 1.40, $-0.12 \pm 1.59i$. No root is expressible by arithmetic and radicals over $\mathbb{Q}$ (a cited result from algebra, beyond this course); the defining polynomial still determines them exactly.

Proposition (Degree one and scalar operators)

On $V \neq \{0\}$, $\deg m_T = 1$ iff $T = cI$ for some $c \in F$; then $m_T = x - c$.

Proof. 1. A monic degree-one relation $m_T = x - c$ says $T - cI = 0$.

2. Conversely, $x - c$ is zero on cI, while a nonzero constant gives a nonzero scalar multiple of I on a nonzero space. Thus no degree-zero polynomial works.

□

Example (A vector relation that is not global)

Over $\mathbb{R}$, let $A = \begin{pmatrix} 2&0&0 \\ 1&2&0 \\ 0&0&3 \end{pmatrix}$.

1. $(A - 2I)e_1 = e_2$ and $(A - 2I)e_2 = 0$. Thus $p_{e_1} = (x-2)^2$: the quadratic works, and $e_1, Ae_1 = (2, 1, 0)$ are independent, excluding a linear relation. Also $p_{e_2} = x - 2$ and $p_{e_3} = x - 3$ by their eigenvector equations.
2. The basis criterion gives $m_A = (x-2)^2(x-3)$. Using only e_1 would have missed the required root 3. To compute the least common multiple here, keep each distinct factor with its largest required exponent; the factor identity and successive cancellation justify this for split polynomials.

Proposition (Direct summands and matrix blocks)

If $V = U_1 \oplus \cdots \oplus U_r$ with each U_j invariant, then $m_T = \operatorname{lcm}(m_{T|_{U_1}}, \ldots, m_{T|_{U_r}})$. Thus a block diagonal matrix, with square blocks on the diagonal and zero off-diagonal blocks, has the least common multiple of the block minimal polynomials. For $A = \begin{pmatrix} B & E \\ 0 & C \end{pmatrix}$, with B, C square, put $r = m_B$, $s = m_C$. Then

$$\operatorname{lcm}(r, s) \mid m_A \mid rs.$$

If $\operatorname{lcm}(r, s) = rs$, both bounds give $m_A = rs$.

Proof. 1. For $v = \sum_j u_j$, $q(T)v = \sum_j q(T|_{U_j})u_j$. Thus $q(T) = 0$ iff $q(T|_{U_j}) = 0$ on every summand, iff every block minimal polynomial divides q. The least-common-multiple definition proves the first assertion.

2. In the upper-block case, the first coordinates form an invariant subspace on which A acts by B; the quotient, identified by the remaining coordinates, acts by C. Relations on invariant parts give $r, s \mid m_A$ and $m_A \mid rs$. The defining property of the least common multiple gives the lower bound.

□

Example (Midterm 2 Review, Problem 5: eigenvalue and relation parts)

Let $B = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ over $\mathbb{R}$.

1. The top-left block has minimal polynomial $(x-3)^2$ because subtracting $3I$ gives a nonzero matrix with square zero. The last block has minimal polynomial $x-1$. These have no shared root, so their least common multiple is their product; the block bounds give $m_B = (x-3)^2(x-1)$.
2. Triangularity gives eigenvalues $3, 1$. For $\lambda = 3$, the equations are $y+z=0, -z=0, -2z=0$, giving $E(3,B) = \mathbb{R}e_1$. For $\lambda = 1$, they are $2x+y+z=0, 2y-z=0$, giving $E(1,B) = \mathbb{R}(-3,2,4)$.
3. The eigenspace dimensions total $2 < 3$, so B is not diagonalizable.

Proposition (Two-by-two relation; LADR 5B Exercise 11)

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$A^2 - (a+d)A + (ad-bc)I = 0.$$

If A is nonscalar, this monic quadratic is m_A; if $A = \gamma I$, then $m_A = x - \gamma$.

Proof. 1. Multiplication gives $A^2 = \left(\begin{smallmatrix} a^2+bc & b(a+d) \\ c(a+d) & d^2+bc \end{smallmatrix}\right)$. Subtracting $(a+d)A$ leaves $(bc-ad)I$, proving the relation directly in entries.

2. A nonscalar matrix has no degree-one relation by the scalar criterion, so this degree-two relation is minimal. This uses no general determinant theorem.

□

Example (A relation from a rank-one product)

For nonzero columns $u, v \in F^n$, $A = uv^{\mathsf{T}}$ has every column a multiple of u, and $A^2 = u(v^{\mathsf{T}}u)v^{\mathsf{T}} = (v^{\mathsf{T}}u)A$. For $u = (1,2)^{\mathsf{T}}$, $v = (3,0)^{\mathsf{T}}$ over $\mathbb{R}$, this gives $A = \left(\begin{smallmatrix} 3 & 0 \\ 6 & 0 \end{smallmatrix}\right)$ and $A^2 = 3A$. Since A is nonscalar, $m_A = x(x-3)$. The factorization of the matrix itself suggested the relation.

Proposition (Reduction to low-degree polynomials)

Every polynomial value $f(T)$ has a unique expression $r(T)$ with $r = 0$ or $\deg r < \deg m_T$. In particular, divide $x^k = m_T h + r$ to compute $T^k = r(T)$.

Proof. 1. Polynomial division gives $f = m_T h + r$, so $f(T) = r(T)$ by $m_T(T) = 0$.

2. If two such remainders give the same operator, their difference is zero on T and has degree below $\deg m_T$ if nonzero. Minimality excludes that possibility, proving uniqueness. On the zero space the only allowed remainder is zero.

□

Compute a high power from a remainder

1. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ over $\mathbb{R}$, $m_A = (x-1)(x-3)$. The remainder of x^k has form $ax+b$. Evaluating the division identity at $1, 3$ gives $a+b=1$ and $3a+b=3^k$.
2. Hence $a = (3^k-1)/2$, $b = (3-3^k)/2$. For $k=4$, $r = 40x - 39$ and $A^4 = 40A - 39I = \left(\begin{smallmatrix} 41 & 40 \\ 40 & 41 \end{smallmatrix}\right)$.

Proposition (Invertibility and a polynomial inverse; LADR 5.32)

T is invertible iff $m_T(0) \neq 0$. For $\dim V > 0$, write $m_T = \sum_{j=0}^{d} a_j x^j$, $a_d = 1$. When $a_0 \neq 0$,

$$T^{-1} = -a_0^{-1} \sum_{j=1}^{d} a_j T^{j-1}.$$

On the zero space $m_T = 1$ and its unique operator is its own inverse.

Proof. 1. By Proposition 66, $m_T(0) = 0$ iff 0 is an eigenvalue, iff T is not invertible by the shifted-kernel criterion (Week 6).

2. Rearrange $\sum_{j=1}^{d} a_j T^j = -a_0 I$, and factor out T. Dividing by $-a_0$ gives a right inverse; it is a polynomial in T, so it commutes with T and is also a left inverse.

□

Compute an inverse from a relation

1. For $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ over $\mathbb{R}$, multiplication gives $A^2 - 5A + 6I = 0$. Thus $A(5I - A) = 6I$.
2. Therefore $A^{-1} = (5I - A)/6 = \left(\begin{smallmatrix} 1/2 & -1/6 \\ 0 & 1/3 \end{smallmatrix}\right)$. Multiplying by A gives $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$: the off-diagonal entry is $-2/6 + 1/3 = 0$.

Changing coordinates or transposing preserves relations.

1. For invertible S, induction gives $(S^{-1}AS)^k = S^{-1}A^kS$, so $p(S^{-1}AS) = S^{-1}p(A)S$. This is zero iff $p(A) = 0$, hence $m_{S^{-1}AS} = m_A$.
2. $(A^{\mathsf{T}})^k = (A^k)^{\mathsf{T}}$, so $p(A^{\mathsf{T}}) = p(A)^{\mathsf{T}}$ is zero iff $p(A) = 0$, giving $m_{A^{\mathsf{T}}} = m_A$.

Divisors of split polynomials. A monic divisor of $\prod_j (x - \lambda_j)$ is a product of some of these factors, with no factor used more often than it appears in the product.

1. Write $ph = \prod_j (x - \lambda_j)$. At λ_1, either $p(\lambda_1) = 0$ or $h(\lambda_1) = 0$. The root factor identity removes $x - \lambda_1$ from that factor; cancel it from both sides.
2. Continue with the remaining linear factors. At the end p is reduced to a constant; its removed factors give its factorization, and monicity makes the constant 1. Distinct factors in the original product therefore remain distinct in a divisor.

Proposition (Compare triangular and diagonal forms; LADR 5.40, 5.44)

T has an upper-triangular matrix over F iff m_T splits into linear factors over F; the factors may repeat. The empty basis and polynomial 1 handle the zero space.

Proof. 1. In a triangular basis with diagonal λ_j, put $U_j = \operatorname{span}(v_1, \ldots, v_j)$, $U_0 = \{0\}$. The column condition gives $(T - \lambda_j I)U_j \subseteq U_{j-1}$. Induction on j shows $\prod_{i=1}^{j}(T - \lambda_i I)$ is zero on U_j: the last factor sends into U_{j-1} and the preceding product is zero there. Thus $\prod_{i=1}^{n}(x - \lambda_i)$ is a global relation. Its divisor m_T splits by the preceding argument.

2. Conversely induct on $\dim V$, starting with the empty basis. For nonzero V with split m_T, a root λ is an eigenvalue by Proposition 66. The proper invariant subspace $U = \operatorname{range}(T - \lambda I)$ has $m_{T|_U} \mid m_T$, hence its polynomial splits too. Induction supplies a triangular basis of U.

3. Extend it by w_j to a basis of V. Each new vector satisfies $Tw_j = (T - \lambda I)w_j + \lambda w_j \in U + Fw_j$. Along with the initial-span condition inside U, this is precisely Proposition 59's triangular column condition.

□

Shortcuts

- A cyclic vector turns minimal-polynomial computation into one vector relation, because the relation propagates to its basis iterates.
- For inverses, isolate the nonzero constant term in $m_T(T) = 0$.
- A root must belong to the chosen field to be an eigenvalue; apply Proposition 66 over that field.

Traps

- An arbitrary annihilating polynomial can have extra roots: $(x-1)(x-2)$ annihilates I, but 2 is not its eigenvalue.
- A relation on one vector need not be a global relation: for $T = \operatorname{diag}(1,2)$, $(T-I)e_1 = 0$ but $T - I \neq 0$.
- The zero-space convention matters: its minimal polynomial is 1, rather than $x - 1$.

§5 Distinct roots and diagonalization

The question. Can a polynomial relation certify an eigenbasis without finding eigenvectors? If $T^2 = I$ over $\mathbb{C}$, then the annihilating polynomial $(x-1)(x+1)$ has distinct roots.

Proposition (Notes Proposition 68; LADR 5.62)

An operator is diagonalizable over F if and only if its minimal polynomial splits into distinct linear factors over F. For the zero space, the empty product 1 satisfies this criterion.

Where the proof comes from

Want. Convert a polynomial relation into an eigenspace decomposition.

Know. Proposition 65 characterizes annihilating polynomials and Proposition 62 tests diagonalization.

Key step. In $m_T(T) = 0$, omit $T - \lambda_j I$ from the product. The remaining product sends every vector into $E(\lambda_j, T)$, because multiplying its output by $T - \lambda_j I$ gives zero. Divide by its nonzero scalar value at λ_j so it is identity on that eigenspace. The normalized polynomials add to 1, giving the full decomposition.

Proof. 1. The zero space has an empty eigenbasis and $m_T = 1$, so assume $V \neq \{0\}$. If T has an eigenbasis, let $\lambda_1, \dots, \lambda_r$ be its distinct eigenvalues. The product $p = \prod_j (x - \lambda_j)$ annihilates every eigenvector, hence T. Thus $m_T \mid p$ by Proposition 65, and Proposition 66 ensures all these roots occur in m_T; monicity gives $m_T = p$.

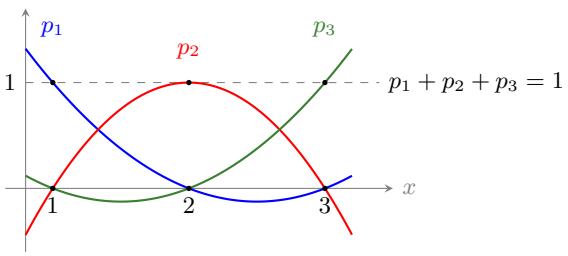
2. Conversely suppose $m_T = \prod_j (x - \lambda_j)$ with distinct roots. Define $p_j(x) = \prod_{k \neq j} (x - \lambda_k)/(\lambda_j - \lambda_k)$.

3. The polynomial $\sum_j p_j - 1$ has degree at most $r - 1$ and vanishes at the r roots, so it is zero by the Week 4 root bound. Hence $\sum_j p_j(T) = I$.

4. Since $(x - \lambda_j)p_j$ is a scalar multiple of m_T, its evaluation is zero. Thus $p_j(T)v \in E(\lambda_j, T)$ for each v.

5. Every $v = \sum_j p_j(T)v$ therefore belongs to the sum of eigenspaces. Proposition 62 proves diagonalizability. The zero-space case follows from its empty eigenbasis.

□



For $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$, each polynomial is 1 at its corresponding eigenvalue and 0 at the other two. Their sum is 1 everywhere. Substituting T therefore decomposes $v = \sum_j p_j(T)v$ into eigenspace components.

Example (Compute the eigenspace components)

For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ over $\mathbb{R}$, $m_A = (x-1)(x-3)$. Thus $p_1 = (3-x)/2$, $p_3 = (x-1)/2$. For $v = (2,0)^{\mathsf{T}}$,

$$p_1(A)v = \tfrac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad p_3(A)v = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The two outputs add to v and satisfy $A(1,-1) = (1,-1)$ and $A(1,1) = 3(1,1)$. More generally, $p_j(T)$ is identity on its eigenspace and zero on the others, because $p_j(T)u = p_j(\lambda_k)u$ for $u \in E(\lambda_k, T)$. This is the projection onto that eigenspace along the others: it returns exactly that component. Hence $p_j(T)^2 = p_j(T)$, $p_i(T)p_j(T) = 0$ for $i \neq j$, and $\sum_j p_j(T) = I$.

Definition (Notes Chapter 13: repeated roots and greatest common divisor)**.** A root λ has multiplicity k if $p = (x-\lambda)^k q$ and $q(\lambda) \neq 0$. A polynomial is squarefree if no root is repeated over $\mathbb{C}$. For $p = \sum_j a_j x^j$, its formal derivative is $p' = \sum_{j \geq 1} j a_j x^{j-1}$. The monic greatest common divisor $\gcd(p,q)$ divides both polynomials and is divisible by every common divisor.

Proposition (Notes Chapter 13: derivative test)

For a nonconstant complex polynomial p, its roots are distinct if and only if $\gcd(p,p') = 1$. Thus over $\mathbb{C}$, T is diagonalizable exactly when $\gcd(m_T, m_T') = 1$ (with the constant case automatic).

Where the proof comes from

Want. Test repeated roots without solving the polynomial.

Know. Week 6 factors complex polynomials into linear factors; division gives the Euclidean algorithm.

Key step. Differentiate a root factorization and track whether the root remains.

Proof. 1. If $p = (x-\lambda)^k q$ with $q(\lambda) \neq 0$, then $p' = k(x-\lambda)^{k-1}q + (x-\lambda)^k q'$. For $k = 1$, $p'(\lambda) = q(\lambda) \neq 0$; for $k > 1$, $p'(\lambda) = 0$.

2. Hence common roots of p, p' are exactly repeated roots. A nonconstant common factor has a root by Week 6, so there are no common roots exactly when the monic gcd is 1.
3. Compute the gcd by repeated polynomial division: replacing (p,q) by $(q, p - hq)$ preserves common divisors; degree decreases until a zero remainder, whose predecessor normalized to be monic is the gcd. Apply Proposition 68.

□

Find a repeated factor by actual polynomial divisions

1. For $p = x^3 - 3x + 2$ over $\mathbb{R}$, $p' = 3x^2 - 3$. The first division is $p = (x/3)p' + (-2x+2)$.
2. The next is $p' = -\frac{3}{2}(x+1)(-2x+2) + 0$. Common divisors are preserved by each remainder step; the last nonzero remainder normalized to be monic is $x - 1$, the greatest common divisor. In particular p has a repeated root at 1: $p(1) = p'(1) = 0$.

Example (Notes Example 59)

Over $\mathbb{C}$, suppose $T^n = I$ for a positive integer n.

1. Proposition 65 gives $m_T \mid x^n - 1$. The derivative of $x^n - 1$ is nx^{n-1}; its only possible root is 0, which is not a root of $x^n - 1$.
2. The derivative test therefore gives distinct roots for $x^n - 1$, hence for its divisor m_T. Proposition 68 gives an eigenbasis.

Example (Notes Example 60; LADR 5.65)

If T is diagonalizable and U invariant, then $T|_U$ and T/U are diagonalizable.

1. Their minimal polynomials divide m_T, by the restriction and quotient result already proved. Proposition 68 gives distinct linear factors for m_T.
2. Each divisor is again a product of distinct linear factors (remove a subset of those factors). Apply Proposition 68 on each space.

Apply the derivative test to a companion matrix (LADR 5.60)

1. For the previous companion matrix, $p = x^5 - 6x + 3$ and $p' = 5x^4 - 6$. If $p(z) = p'(z) = 0$, then $z^4 = 6/5$ and $p(z) = -(24/5)z + 3 = 0$, so $z = 5/8$.
2. But $(5/8)^4 = 625/4096 \neq 6/5$ since $3125 \neq 24576$. Thus no common root exists, and the matrix is diagonalizable over $\mathbb{C}$. This gives an exact justification without relying on approximate roots.

Detect a repeated factor (LADR 5.61)

1. Over $\mathbb{R}$ or $\mathbb{C}$, for $B = \left(\begin{smallmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{smallmatrix}\right)$, the eigenvalues are 6 and 7. Proposition 67 bounds $\deg m_B \leq 3$, and Proposition 66 requires both roots.
2. Compute $(B - 6I)(B - 7I) = \left(\begin{smallmatrix} 0 & -3 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) \neq 0$, whereas $(B - 6I)^2(B - 7I) = 0$. Hence Proposition 65 gives $m_B = (x - 6)^2(x - 7)$, and Proposition 68 rules out diagonalization.

Shortcuts

- Use the Euclidean algorithm on m_T, m_T' over $\mathbb{C}$ to avoid computing roots.
- If an annihilating polynomial splits into distinct factors, so does m_T by divisibility, immediately proving diagonalizability.

Traps

- Squarefree alone is insufficient over $\mathbb{R}$: $x^2 + 1$ is squarefree but does not split, and the quarter-turn is not diagonalizable over $\mathbb{R}$.
- An annihilating polynomial with repeated roots does not rule out an eigenbasis: $(x - 1)^2$ annihilates I. The criterion concerns the minimal polynomial.

§6 Further eigenvalue tests from the textbook

The question. Can one guarantee or locate eigenvalues without solving their polynomial? Every real operator in odd dimension has a real eigenvalue, and a matrix's row sizes bound where its eigenvalues can lie.

Proposition (LADR 5.33: even quadratic kernel)

For an operator T on a finite-dimensional real space and real b, c with $b^2 < 4c$, the dimension of $\ker(T^2 + bT + cI)$ is even.

Where the proof comes from

Want. Prove a parity restriction using invariant subspaces.

Know. $x^2 + bx + c = (x + b/2)^2 + c - b^2/4$ has no real root.

Key step. The quadratic relation gives $T^2 w = -bTw - cw$, so $\operatorname{span}(w, Tw)$ is invariant. No real root means no invariant line; this span has dimension 2. If an even-dimensional invariant subspace is not all of the kernel, adjoining such a span adds two dimensions: an intersection of dimension 1 would be an invariant line. Repeat until the kernel is exhausted.

Proof. 1. Restrict T to $K = \ker(T^2 + bT + cI)$, invariant by Week 6. On K, $T^2 + bT + cI = 0$, so no nonzero vector is a real eigenvector, by substitution and positivity of the quadratic.
2. Choose an invariant subspace $U \subseteq K$ of largest even dimension; 0 is available and dimensions are bounded integers. If $w \in K \setminus U$, the space $W = \operatorname{span}(w, Tw)$ is invariant because $T^2 w = -bTw - cw$, and has dimension 2 because w is not an eigenvector.
3. Since $T(U \cap W) \subseteq T(U) \cap T(W) \subseteq U \cap W$, the intersection is invariant. It cannot have dimension 1 (that would be an invariant line), or 2 (then $w \in U$). Thus it is zero.
4. Then $U + W$ is invariant of even dimension $\dim U + 2$, contradicting maximality. Therefore $U = K$ and $\dim K$ is even.

□

Proposition (Real polynomial factorization; LADR 4.16)

Every nonconstant real polynomial is a product of a nonzero real constant, real linear factors, and real quadratic factors with negative discriminant.

Proof. 1. For real coefficients, conjugating the finite sum gives $\overline{p(z)} = p(\overline{z})$. Thus nonreal roots occur in conjugate pairs, with the same multiplicity, by conjugating their factorization.
2. For $z = a + bi$ with $b \neq 0$, the pair contributes $(x - z)(x - \overline{z}) = x^2 - 2ax + a^2 + b^2$, whose discriminant is $-4b^2 < 0$.
3. Group these pairs and retain the real roots as linear factors in the complex factorization from Week 6. The leading real coefficient is the remaining constant.

□

Proposition (LADR 5.34: odd-dimensional real operators)

Every operator on a real vector space of positive odd finite dimension has an eigenvalue.

Where the proof comes from

Want. Force a real eigenvalue even though a real polynomial need not have a real root.

Know. Real factorization gives linear or irreducible quadratic factors; the preceding result gives even quadratic-kernel dimension.

Key step. Choose the range of the quadratic factor: rank–nullity makes its dimension the odd number $\dim V$ minus an even kernel dimension. Minimality makes that range proper, since an onto quadratic factor would leave a smaller polynomial relation on all of V. It is therefore a smaller positive odd-dimensional invariant space where induction applies.

Proof. 1. Induct on odd n. For $n = 1$, every operator is scalar. If m_T has a real linear factor, Proposition 66 gives an eigenvalue.
2. Otherwise real factorization supplies $m_T = qh$ with $q = x^2 + bx + c$, $b^2 < 4c$. Put $W = \operatorname{range} q(T)$, invariant by Week 6.
3. If $W = V$, $h(T)q(T) = 0$ would imply $h(T) = 0$, contradicting minimality. Thus W is proper.
4. Rank-nullity and the even-kernel result give $\dim W = n - \dim \ker q(T)$ odd, hence positive. It is less than n, so induction gives a real eigenvector of $T|_W$, also one of T.

□

Definition (LADR 5.66: Gershgorin disks)**.** For $F = \mathbb{R}$ or $\mathbb{C}$ and a matrix $A \in F^{n \times n}$, its ith Gershgorin disk is $D_i = \{z \in F : |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|\}$. Over $\mathbb{R}$ these sets are intervals; to locate all complex eigenvalues of a real matrix, regard that matrix over $\mathbb{C}$.

Proposition (LADR 5.67: Gershgorin disk theorem)

Every eigenvalue of A lies in $\bigcup_i D_i$.

Where the proof comes from

Want. Locate eigenvalues from row entries.

Know. An eigenvector has a coordinate of maximum nonzero modulus.

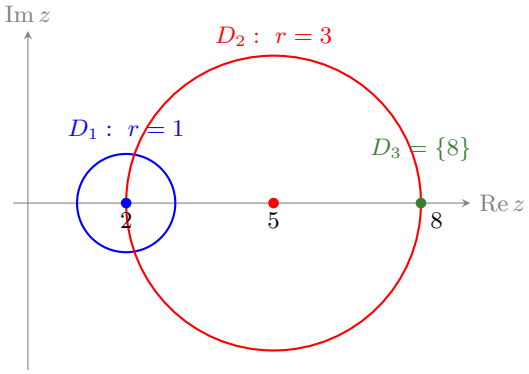
Key step. The row equation contains unknown ratios x_j/x_i. Choose $|x_i|$ largest so each ratio has modulus at most 1; dividing by this nonzero coordinate then leaves a bound involving only the row entries.

Proof. 1. Choose an eigenvector $x \neq 0$ and an index i with $|x_i| = \max_j |x_j| > 0$. The ith row gives $(\lambda - a_{ii})x_i = \sum_{j \neq i} a_{ij} x_j$.
2. The triangle inequality yields $|\lambda - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|\,|x_j|/|x_i| \leq \sum_{j \neq i} |a_{ij}|$. Thus $\lambda \in D_i$.

□

Bound eigenvalues by row sizes

1. For $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 5 & 3 \\ 0 & 0 & 8 \end{smallmatrix}\right)$, the three disks have centers $2, 5, 8$ and radii $1, 3, 0$. Thus each eigenvalue belongs to $\{|z-2| \leq 1\} \cup \{|z-5| \leq 3\} \cup \{8\}$.
2. The triangular test gives the actual eigenvalues $2, 5, 8$, and each lies in the stated union. Gershgorin provides bounds for matrices that are not triangular as well.



The row bounds for the displayed matrix give disks centered at $2, 5, 8$ with radii $1, 3, 0$. All three actual eigenvalues lie in the union. Most points of the disks are not eigenvalues; the theorem gives a region to search.

Shortcuts

- A real odd-dimensional operator has a real eigenvalue without any root calculation, by the parity proof.
- Gershgorin bounds require only diagonal entries and sums of off-diagonal absolute values.

Traps

- Even real dimension need not give a real eigenvalue: the quarter-turn on $\mathbb{R}^2$ has none.
- Disk membership alone does not characterize eigenvalues: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has unit disks containing 0, but its only eigenvalues are ± 1.

§7 Problem set

HW7 P1. Let $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. (a) Find m_A over $\mathbb{R}$ and $\mathbb{C}$. (b) Find m_B over $\mathbb{R}$. (c) Recover the diagonalizability conclusions of HW6 using Proposition 68.
Hint: Use the first-relation method: square A and $B - I$, then rule out scalar matrices.

HW7 P2. For differentiation $D : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, find its minimal polynomial and decide diagonalizability.
Hint: Use the first-relation method on the iterates of x^3.

HW7 P3. Let T be invertible on a finite-dimensional space and $m_T = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$. (a) Prove $a_0 \neq 0$. (b) Prove T^{-1} is a polynomial in T. (c) Over $\mathbb{C}$, if $m_T = x^4 + x^3 - x^2 - 5x + 4$, express T^{-1} explicitly, find the eigenvalues, and decide diagonalizability.
Hint: Use LADR 5.32 to isolate $a_0 I$; apply Propositions 66 and 68 after factoring.

HW7 P4 (optional). Prove $m_{T-\lambda I}(x) = m_T(x + \lambda)$. If $c \neq 0$ and $d = \deg m_T$, prove $m_{cT}(x) = c^d m_T(x/c)$.
Hint: Use the minimality definition after affine substitution, then reverse the substitution to exclude smaller degree.

HW7 P5. Let A be a two-by-two complex matrix with only eigenvalue λ. Prove m_A is either $x - \lambda$ or $(x - \lambda)^2$; prove the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable exactly in that case.
Hint: Combine complex factorization, Propositions 66–68, and the degree bound.

HW7 P6. For $C = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, find eigenvalues, eigenspaces and minimal polynomial, and decide diagonalizability.
Hint: Use Proposition 60 and kernel equations, then Proposition 65 to test $(C - 2I)(C - 3I)$.

HW7 P7 (optional). Suppose $m_T = \prod_{j=1}^{r}(x - \lambda_j)$ with distinct roots. Construct p_j with $p_j(\lambda_j) = 1$, $p_j(\lambda_k) = 0$ for $k \neq j$, and show that $p_j(T)$ projects onto $E(\lambda_j, T)$ along the other eigenspaces.
Hint: Use the interpolation polynomials in the proof of Proposition 68.

Exam practice (midterm2-review.pdf, Problem 6; Axler 5C.7). For T on finite-dimensional V and $v \in V$, prove that there is a unique monic polynomial p_v of smallest degree with $p_v(T)v = 0$, and prove $p_v \mid m_T$.
Hint: Use the first-dependence argument and polynomial division on m_T by p_v; when $v = 0$, use $p_v = 1$.

Actual Midterm 2, October 30, Question 3. On $\mathbb{R}^3$, let T have standard matrix $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right)$. (a) Determine its eigenvalues and corresponding eigenspaces. (b) Determine its minimal polynomial with justification. (c) Is T diagonalizable? Explain.
Hint: Use the shifted-kernel equations and test both the quadratic and cubic products; roots alone do not justify the exponent.

Practice Midterm 2, Question 3. Let T act on a nonzero finite-dimensional complex space. You may use eigenvalue existence. Prove upper-triangularization by induction: (a) handle dimension 1; (b) choose an eigenvalue λ, put $U = \operatorname{range}(T - \lambda I)$, and prove invariance and $\dim U < \dim V$; (c) obtain a basis $u_1, \ldots, u_m$ of U with $Tu_j \in \operatorname{span}(u_1, \ldots, u_j)$; (d) extend it by $v_1, \ldots, v_r$ and use $Tv_j = (T - \lambda I)v_j + \lambda v_j$.
Hint: Proposition 61 explains why the range is the useful smaller space; include the case $U = \{0\}$.

Practice Midterm 2, Question 4. Let T act on a nonzero finite-dimensional real space. (a) If $m_T(0) \neq 0$, prove invertibility by expressing T^{-1} as a polynomial. (b) If $m_T(0) = 0$, prove noninvertibility using minimality. (c) For $R(x, y) = (-y, x)$ on $\mathbb{R}^2$, find m_R. (d) Use (a) to express R^{-1} in terms of R and check directly.
Hint: Factor out T from every nonconstant term; in (b), an inverse would cancel a factor x and produce a smaller relation.

Midterm 2 Review, Problem 4. (1) On the real space $V = \operatorname{span}(1, \sin x, \cos x, \sin 2x, \cos 2x)$, find the eigenvalues of differentiation and decide diagonalizability. (2) For $T(x, y) = (y, -x)$ on $\mathbb{C}^2$, find eigenvalues and a basis giving an upper-triangular matrix. (3) Show the same rule on $\mathbb{R}^2$ has no upper-triangular basis.
Hint: Write the derivative equations in the displayed function basis; in (2),(3), check the field and the sign of the rotation.